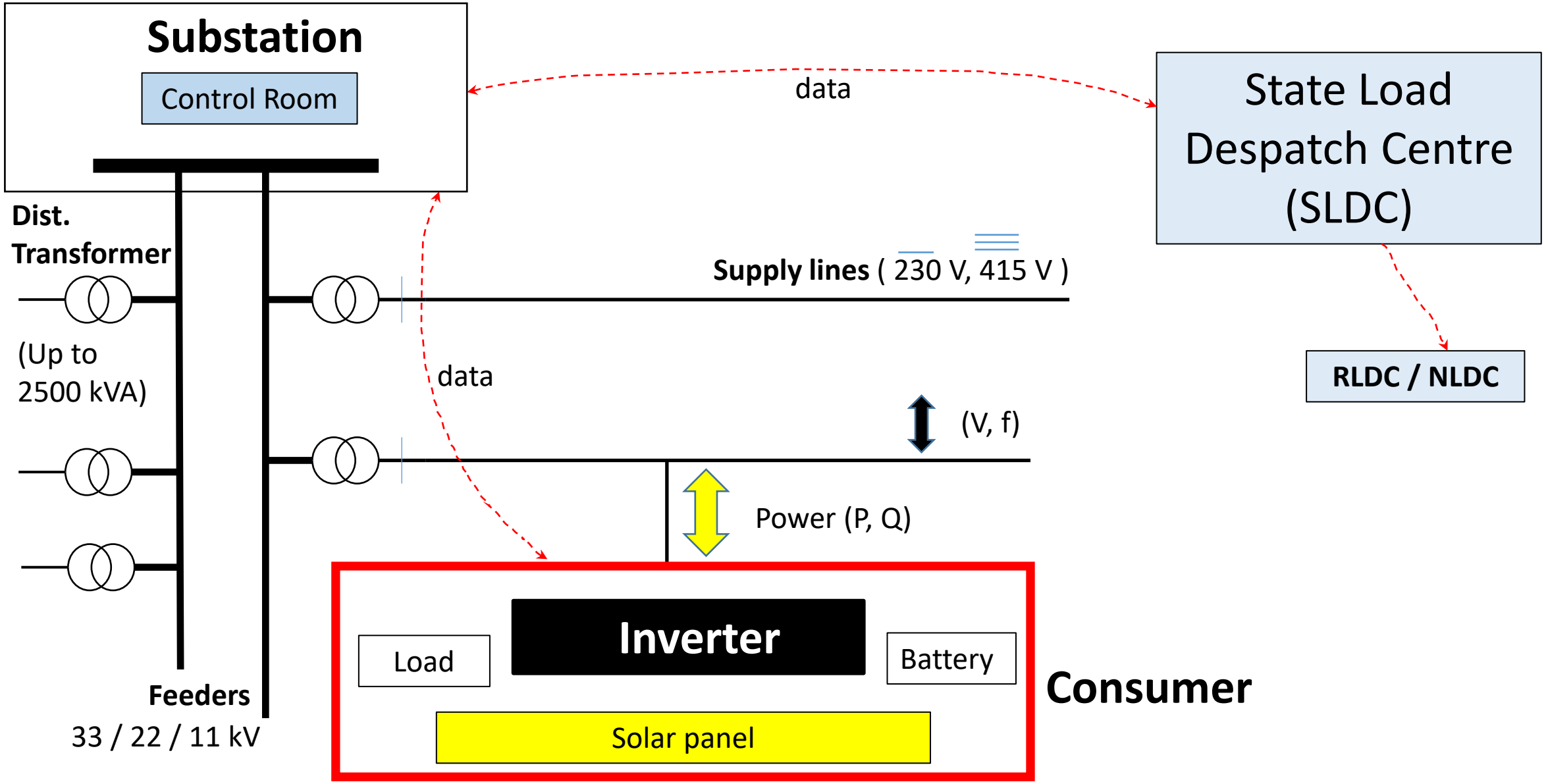
Smart Inverters

Distribution System



Smart Inverters

- **Conventional Inverters:**

  merely convert power and shut down during disturbances.

- **Smart Inverters**:

  - convert power
  - communicate with the grid
  - respond autonomously to voltage (V) and frequency (f) fluctuations
  - manage grid stability, safety, and efficiency through automated functionalities.

- **Key aspects of Smart Inverters:**
  - **Standards & compliance (BIS/BEE):**
    - must comply with BIS and MNRE guidelines (for reliability and grid safety)
    - **IS 17980:2022** (for MPPT efficiency)
    - **IS 16221-2:2015** (for safety of power converters)
  - **Grid stability support functions (autonomous control):**
    - **Voltage and Frequency Ride-through:**
      - ability to remain connected during brief, minor disturbances.
    - **Dynamic reactive (Q) and real (P) power support:**
      - adjusting output to support grid stability.
    - **Ramp rate controls:**
      - managing the speed at which power output changes.

Smart Inverters (contd…)

- **Key aspects of Smart Inverters (contd…)**
  - **Intelligent Monitoring and Connectivity (IoT):** capabilities Internet of Things (IoT)
    - **Real-time Monitoring:**
      - tracking performance via smartphone.
    - **Remote Diagnostics:**
      - allowing technicians to diagnose issues remotely.
    - **Anti-Islanding Protection:**
      - ensuring safe, automatic shutdown during grid failure to protect utility workers.
  - **Battery Integration and Energy Management:**
    - **optimizing** energy flow between solar panels, battery storage and the grid.
    - **designed** for high efficiency, employing MPPT to maximize yield.
  - **Mandatory standards from January 1, 2026:**
    - BEE's **Standards & Labeling (S&L)**
      - for grid-connected solar inverters up to **100 kW,**
      - with specific minimum efficiency requirements (e.g. 98% for >20 kW)

Table 1: Minimum Energy Performance Standard for Grid Connected Solar Inverter

Rated Output Power (kW)	Minimum Overall Efficiency Requirement
Rated Output Power < 1	92%
1 ≤ Rated Output Power < 3	93%
3 ≤ Rated Output Power < 5	95%
5 ≤ Rated Output Power < 10	96%
10 ≤ Rated Output Power < 20	97%
Rated Output Power ≥ 20	98%

Challenges

- Power quality issues:
  - Voltage fluctuations
  - Harmonics
- Reverse power flow
- Digital tools for monitoring & maintaining technical attributes
- Cyber Security: data breach; device control.

Thank you!